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PROJECT REPORT

Stability and Bifurcation of a Simple Food Chain in a Chemostat
with Removal Rates

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2 Mathematical Model and Theory

The reference first formulates the food chain in dimensional variables and then rescales concentration and time. Nutrient concentration is measured relative to the input concentration, time is measured in units of the inverse chemostat dilution rate, and the organism concentrations are scaled using the corresponding yield constants. This nondimensionalization fixes the nutrient input and dilution terms at one and reduces the model to four state variables and three effective removal rates. The model used in the analysis and in our implementation is

$$\begin{aligned}\dot{S} &= 1 - S - f_1(S)x, \\ \dot{x} &= x(f_1(S) - D_1) - f_2(x)y, \\ \dot{y} &= y(f_2(x) - D_2) - f_3(y)z, \\ \dot{z} &= z(f_3(y) - D_3).\end{aligned}\tag{1}$$

Here and below a dot denotes differentiation with respect to dimensionless time.

2.1 State Variables and Model Parameters

The variables form the state vector $X(t) = (S(t), x(t), y(t), z(t))^T \in \mathbb{R}_+^4$. Their meanings are summarized in Table 1.

Table 1: State variables of the nondimensional chemostat model.

Symbol	Biological meaning	Domain
$S(t)$	Nutrient concentration	$S \geq 0$
$x(t)$	Prey concentration; consumes the nutrient	$x \geq 0$
$y(t)$	First-predator concentration; consumes x	$y \geq 0$
$z(t)$	Top-predator concentration; consumes y	$z \geq 0$

The paper allows general response functions f_i . It assumes that every $f_i : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuously differentiable, $f_i(0) = 0$, and $f'_i(u) > 0$ for $u > 0$. Thus, growth increases with the available food concentration. For reproducible simulations, the project specializes these functions to the Monod form

$$f_i(u) = \frac{a_i u}{k_i + u}, \quad a_i > 0, \quad k_i > 0, \quad i = 1, 2, 3.\tag{2}$$

The parameter a_i is the maximum specific growth rate and k_i is the half-saturation constant, since $f_i(k_i) = a_i/2$. The derivative and inverse needed in the equilibrium calculations are

$$f'_i(u) = \frac{a_i k_i}{(k_i + u)^2}, \quad f_i^{-1}(d) = \frac{dk_i}{a_i - d} \quad (0 \leq d < a_i).\tag{3}$$

Each $D_i > 0$ is the dimensionless total removal rate of one organism. In the dimensional model this rate is the sum of chemostat washout and the organism's specific death rate; division by the dilution rate produces the dimensionless D_i . The nine numerical parameters used by the simulator are therefore

$$(a_1, k_1, a_2, k_2, a_3, k_3, D_1, D_2, D_3).$$

2.2 Governing Differential Equations

The first equation of (1) balances normalized nutrient inflow, nutrient washout, and consumption by the prey. The prey grows at per-capita rate $f_1(S)$, is removed at rate D_1 , and is consumed by the first predator through the flux $f_2(x)y$. Similarly, y grows through prey consumption, is

removed at rate D_2 , and is consumed by z . The top predator grows at rate $f_3(y)$ and is removed at rate D_3 .

This chain structure is important: y does not consume nutrient directly, and z consumes neither nutrient nor prey. The transfer terms cancel when all four equations are added. Distinct removal rates prevent the stronger conservation law available in equal-removal chemostat models, so the complete four-dimensional system must be studied.

2.3 Initial Conditions and Parameter Restrictions

The reference considers strictly positive initial data,

$$S(0) = S_0 > 0, \quad x(0) = x_0 > 0, \quad y(0) = y_0 > 0, \quad z(0) = z_0 > 0. \quad (4)$$

Boundary initial states are also meaningful when studying washout equilibria. All model parameters are positive. In addition, a finite positive solution of $f_i(u) = D_i$ requires $D_i < a_i$ for the Monod functions. Stronger conditions are needed for an equilibrium to lie in the biologically feasible region; these are derived below rather than imposed globally.

2.4 Nonnegativity, Boundedness, and Biological Feasibility

The nonnegative orthant is positively invariant. At $S = 0$, the first equation gives $\dot{S} = 1 > 0$. At $x = 0$, $y = 0$, or $z = 0$, the corresponding derivative is zero because $f_i(0) = 0$. Therefore a solution starting with nonnegative concentrations cannot cross into negative values.

To establish boundedness, define the total dimensionless concentration

$$T(t) = S(t) + x(t) + y(t) + z(t).$$

Adding the equations in (1) gives

$$\dot{T} = 1 - S - D_1x - D_2y - D_3z. \quad (5)$$

Let

$$D_{\min} = \min\{1, D_1, D_2, D_3\}, \quad D_{\max} = \max\{1, D_1, D_2, D_3\}.$$

Since all state variables are nonnegative,

$$1 - D_{\max}T \leq \dot{T} \leq 1 - D_{\min}T. \quad (6)$$

The scalar comparison theorem consequently yields

$$\frac{1}{D_{\max}} \leq \liminf_{t \rightarrow \infty} T(t) \leq \limsup_{t \rightarrow \infty} T(t) \leq \frac{1}{D_{\min}}. \quad (7)$$

Equivalently, for any sufficiently small $\kappa > 0$, trajectories are eventually attracted to the bounded region

$$\Gamma_\kappa = \left\{ X \in \mathbb{R}_+^4 : \frac{1}{D_{\max}} - \kappa \leq S + x + y + z \leq \frac{1}{D_{\min}} + \kappa \right\}. \quad (8)$$

Thus the model is dissipative: long-term trajectories enter a common bounded set. This result is both biologically necessary and numerically useful, because an unbounded simulated trajectory would indicate an invalid parameter or integration failure rather than intended model behavior.

2.5 Equilibrium Points

An equilibrium $P = (S^*, x^*, y^*, z^*)$ satisfies

$$\begin{aligned} 1 - S^* - f_1(S^*)x^* &= 0, \\ x^*(f_1(S^*) - D_1) - f_2(x^*)y^* &= 0, \\ y^*(f_2(x^*) - D_2) - f_3(y^*)z^* &= 0, \\ z^*(f_3(y^*) - D_3) &= 0. \end{aligned} \quad (9)$$

The four possible trophic regimes follow by deciding which organism concentrations are zero.

2.5.1 Washout Equilibrium P_0

If all organisms are absent, nutrient converges to its normalized input level:

$$P_0 = (1, 0, 0, 0). \quad (10)$$

This equilibrium always exists.

2.5.2 Prey-Only Equilibrium P_1

For $x_1 > 0$ and $y_1 = z_1 = 0$, the prey equation requires $f_1(S_1) = D_1$. The nutrient balance then gives

$$P_1 = (S_1, x_1, 0, 0), \quad S_1 = f_1^{-1}(D_1), \quad x_1 = \frac{1 - S_1}{D_1}. \quad (11)$$

For Monod growth,

$$S_1 = \frac{D_1 k_1}{a_1 - D_1}. \quad (12)$$

The biological condition $0 < S_1 < 1$ is equivalent to $D_1 < f_1(1) = a_1/(k_1 + 1)$. This is precisely the condition that prey can increase when rare at P_0 .

2.5.3 Two-Species Equilibrium P_2

For prey and the first predator to persist while $z = 0$, the y -equation requires $f_2(x_2) = D_2$. Therefore

$$\begin{aligned} P_2 &= (S_2, x_2, y_2, 0), \\ x_2 &= f_2^{-1}(D_2), \\ S_2 + f_1(S_2)x_2 &= 1, \\ y_2 &= \frac{x_2(f_1(S_2) - D_1)}{D_2}. \end{aligned} \quad (13)$$

For the Monod function, $x_2 = D_2 k_2/(a_2 - D_2)$. Once x_2 is known, the equation for S_2 has a unique root in $(0, 1)$, because $S + f_1(S)x_2$ is strictly increasing from zero to a value greater than one. Finally, $y_2 > 0$ requires $f_1(S_2) > D_1$. In the equilibrium hierarchy, this condition is equivalent to the first-predator invasion threshold $f_2(x_1) > D_2$.

2.5.4 Coexistence Equilibrium P_3

At an interior equilibrium all four components are positive. The top-predator equation first fixes

$$y_3 = f_3^{-1}(D_3) = \frac{D_3 k_3}{a_3 - D_3} \quad (14)$$

for Monod growth. The remaining coordinates satisfy

$$\begin{aligned} P_3 &= (S_3, x_3, y_3, z_3), \\ S_3 + f_1(S_3)x_3 &= 1, \\ x_3(f_1(S_3) - D_1) &= f_2(x_3)y_3, \\ z_3 &= \frac{y_3(f_2(x_3) - D_2)}{D_3}. \end{aligned} \quad (15)$$

The coupled equations for S_3 and x_3 generally do not have a useful closed form, even after choosing Monod functions. The implementation solves $S + f_1(S)x = 1$ by bisection for each trial x , then locates a sign change in the prey-balance residual. The coexistence state is feasible only when all coordinates are positive. In particular, $D_3 < a_3$ is needed to define y_3 , and $f_2(x_3) > D_2$ is needed for $z_3 > 0$.

2.6 Existence and Positivity Conditions

The equilibria form a trophic hierarchy controlled by invasion rates. An invasion rate is the per-capita growth rate of a missing species when that species is introduced at very low concentration. The three relevant rates are

$$\lambda_x = f_1(1) - D_1, \quad \lambda_y = f_2(x_1) - D_2, \quad \lambda_z = f_3(y_2) - D_3. \quad (16)$$

Table 2: Equilibrium hierarchy and biological existence conditions.

Equilibrium	Persisting organisms	Main feasibility condition
P_0	None	Always exists
P_1	x	$\lambda_x > 0$
P_2	x, y	P_1 exists and $\lambda_y > 0$
P_3	x, y, z	P_2 exists and $\lambda_z > 0$

Positive invasion means that the next trophic level can enter the lower-level equilibrium. Negative invasion means that it washes out when rare. At a zero invasion rate the corresponding equilibrium is nonhyperbolic, and a change in the qualitative survival regime may occur.

2.7 Local Stability and Threshold Conditions

Linearizing (1) at a state (S, x, y, z) gives the Jacobian

$$J = \begin{pmatrix} -1 - f'_1(S)x & -f_1(S) & 0 & 0 \\ f'_1(S)x & f_1(S) - D_1 - f'_2(x)y & -f_2(x) & 0 \\ 0 & f'_2(x)y & f_2(x) - D_2 - f'_3(y)z & -f_3(y) \\ 0 & 0 & f'_3(y)z & f_3(y) - D_3 \end{pmatrix}. \quad (17)$$

An equilibrium is locally asymptotically stable when every eigenvalue of its Jacobian has negative real part.

Stability of P_0 . At washout, the eigenvalues are

$$-1, \quad f_1(1) - D_1, \quad -D_2, \quad -D_3. \quad (18)$$

Consequently,

$$P_0 \text{ is locally asymptotically stable} \iff f_1(1) - D_1 < 0. \quad (19)$$

For Monod growth this becomes $a_1/(k_1 + 1) < D_1$. If the inequality is reversed, prey can invade and P_0 is unstable in the x -direction.

Stability of P_1 . The (S, x) block of $J(P_1)$ is

$$J_{Sx}(P_1) = \begin{pmatrix} -1 - x_1 f'_1(S_1) & -D_1 \\ x_1 f'_1(S_1) & 0 \end{pmatrix}.$$

Its trace is negative and its determinant is positive, so both eigenvalues have negative real part. The remaining eigenvalues are $f_2(x_1) - D_2$ and $-D_3$. Hence

$$P_1 \text{ is locally asymptotically stable} \iff f_2(x_1) - D_2 < 0. \quad (20)$$

When this threshold is positive, P_1 is a saddle that repels trajectories in the y -direction.

Stability of P_2 . For a compact Routh–Hurwitz representation, define

$$\begin{aligned} a &= 1 + x_2 f'_1(S_2), & b &= f_1(S_2), & c &= x_2 f'_1(S_2), \\ d &= y_2 f'_2(x_2), & e &= D_2, & r &= f_1(S_2) - D_1 - y_2 f'_2(x_2). \end{aligned}$$

The active (S, x, y) block is

$$\begin{pmatrix} -a & -b & 0 \\ c & r & -e \\ 0 & d & 0 \end{pmatrix},$$

whose characteristic polynomial is

$$\lambda^3 + A_1 \lambda^2 + A_2 \lambda + A_3, \quad (21)$$

with

$$A_1 = a - r, \quad A_2 = ed - ar + bc, \quad A_3 = aed. \quad (22)$$

The fourth eigenvalue is the top-predator invasion rate $\lambda_z = f_3(y_2) - D_3$. The cubic Routh–Hurwitz criterion gives

$$A_1 > 0, \quad A_2 > 0, \quad A_3 > 0, \quad A_1 A_2 > A_3, \quad \text{and} \quad f_3(y_2) - D_3 < 0 \quad (23)$$

as sufficient and necessary local stability conditions. If the active block is stable but $f_3(y_2) - D_3 > 0$, then P_2 is a saddle repelling in the z -direction.

Stability of P_3 . At coexistence, introduce the abbreviations

$$\begin{aligned} A &= 1 + x_3 f'_1(S_3), & B &= f_1(S_3), & C &= x_3 f'_1(S_3), \\ R &= f_1(S_3) - D_1 - y_3 f'_2(x_3), & E &= f_2(x_3), & F &= y_3 f'_2(x_3), \\ Q &= f_2(x_3) - D_2 - z_3 f'_3(y_3), & G &= f_3(y_3), & H &= z_3 f'_3(y_3). \end{aligned}$$

Then

$$J(P_3) = \begin{pmatrix} -A & -B & 0 & 0 \\ C & R & -E & 0 \\ 0 & F & Q & -G \\ 0 & 0 & H & 0 \end{pmatrix}. \quad (24)$$

Set $p = A - R$ and $q = BC - AR$. Expanding $\det(\lambda I - J(P_3))$ gives

$$\lambda^4 + C_1 \lambda^3 + C_2 \lambda^2 + C_3 \lambda + C_4, \quad (25)$$

where

$$\begin{aligned} C_1 &= p - Q, \\ C_2 &= q - Qp + EF + GH, \\ C_3 &= EFA - Qq + GHp, \\ C_4 &= GHq. \end{aligned} \quad (26)$$

The quartic Routh–Hurwitz conditions are

$$\begin{aligned} C_1 &> 0, \quad C_2 > 0, \quad C_3 > 0, \quad C_4 > 0, \\ C_1 C_2 &> C_3, \quad C_1 C_2 C_3 &> C_3^2 + C_1^2 C_4. \end{aligned} \quad (27)$$

These inequalities provide a direct analytical or numerical test for local coexistence stability.

2.8 Persistence and Bifurcation Behavior

Local stability describes trajectories close to one equilibrium, whereas uniform persistence concerns all positive solutions. The reference combines the dissipativity result with boundary-flow arguments. In particular, if P_1 is globally attracting on the (S, x) boundary subsystem but repels the missing first predator, and P_2 is globally attracting on the (S, x, y) boundary subsystem but repels the missing top predator, then no strictly positive trajectory can approach the boundary. Under the paper's additional hyperbolicity and no-cycle assumptions, there is an $\eta > 0$ such that

$$\liminf_{t \rightarrow \infty} S(t), \quad \liminf_{t \rightarrow \infty} x(t), \quad \liminf_{t \rightarrow \infty} y(t), \quad \liminf_{t \rightarrow \infty} z(t) \geq \eta. \quad (28)$$

This uniform persistence implies the existence of at least one positive interior equilibrium P_3 .

For bifurcation analysis, the paper introduces a real parameter μ into the prey growth response, writing $f_1(S; \mu)$, and studies how the Jacobian eigenvalues change with μ . A Hopf bifurcation requires a complex conjugate pair to cross the imaginary axis with nonzero speed while the other eigenvalues remain away from it.

At P_2 , the cubic factor reaches a candidate Hopf boundary when

$$A_1 A_2 = A_3, \quad (29)$$

with positive coefficients, a nonzero top-predator eigenvalue, and the transversality condition satisfied. At P_3 , the corresponding quartic boundary is

$$C_1 C_2 C_3 = C_3^2 + C_1^2 C_4, \quad (30)$$

again together with coefficient positivity and transversality. Crossing one of these boundaries can create a family of periodic solutions. The numerical oscillatory experiment later in this report illustrates behavior near such a boundary, but a damped oscillatory trajectory alone is not a proof of Hopf bifurcation.

2.9 Consistency of the Implemented Model with the Reference Analysis

Two inconsistencies in the printed article must be handled explicitly. First, the displayed nondimensional equation (1.2) appears to repeat population factors inside several per-capita growth terms. Read literally, that display does not produce the paper's later Jacobian or equilibrium formulas. The system in (1) does produce them and is therefore the model used by the simulator, equilibrium calculations, and this report. Second, the final discussion states the opposite sign for washout stability, while Theorem 2.2 and the eigenvalues at P_0 correctly require $f_1(1) - D_1 < 0$. We follow the theorem and the direct linearization.

The code mirrors the mathematical construction as follows:

- `equilibria_stability/model.py` implements (2), its derivative, and its inverse;
- `equilibria_stability/analysis.py` computes P_0 through P_3 , checks positivity, and evaluates the P_0 threshold;
- `demo/app.py` evaluates (17) and computes its characteristic roots for the interactive local-stability display.

Table 3 gives representative outputs produced by the equilibrium implementation. They also serve as analytical targets for the simulations in Section 4.

Table 3: Representative equilibrium calculations from the implemented model.

Scenario	Richest feasible equilibrium	Coordinates (S, x, y, z)
Washout	P_0	$(1, 0, 0, 0)$
Prey only	P_1	$(0.166667, 1.666667, 0, 0)$
Two species	P_2	$(0.628667, 0.333333, 0.409333, 0)$
Coexistence	P_3	$(0.055067, 0.804597, 0.280534, 0.081550)$

4 Simulation Experiments and Results

4.1 Experimental Setup

The purpose of the simulation experiments is to connect the analytical equilibria of the chemostat food-chain model with visible long-time dynamics. All runs use the internally consistent four-dimensional model

$$\begin{aligned}\frac{dS}{dt} &= 1 - S - f_1(S)x, \\ \frac{dx}{dt} &= x(f_1(S) - D_1) - f_2(x)y, \\ \frac{dy}{dt} &= y(f_2(x) - D_2) - f_3(y)z, \\ \frac{dz}{dt} &= z(f_3(y) - D_3),\end{aligned}$$

where S is the nutrient concentration, x is the prey population, y is the first predator, and z is the top predator. The response functions are Monod functions,

$$f_i(u) = \frac{a_i u}{k_i + u}, \quad i = 1, 2, 3.$$

This version of the system is the one used in the equilibrium and Jacobian analysis. It was also used by the simulator so that the numerical experiments and the theoretical formulas describe the same model.

The trajectories were computed with the classical fourth-order Runge–Kutta method using time step $\Delta t = 0.01$. The initial condition was fixed at

$$(S_0, x_0, y_0, z_0) = (0.5, 0.2, 0.1, 0.05)$$

for all five main experiments. The four equilibrium scenarios P_0 through P_3 were integrated until $t = 500$. The oscillatory case was integrated until $t = 2000$, because its transient oscillations decay slowly and a shorter run could give a misleading classification. The simulator wrote every tenth time step to CSV files and recorded diagnostics including final states, componentwise minima and maxima, negative-state clamps, explosion status, and distances from the analytical equilibria. No run exploded and no run required negative-state clamping.

4.2 Parameter-Set Selection

Five parameter sets were selected to represent the main biological regimes predicted by the equilibrium hierarchy: complete washout P_0 , prey-only survival P_1 , survival of the prey and first predator P_2 , coexistence of all trophic levels P_3 , and a slowly damped oscillatory response near an interior equilibrium. The values used in the simulations are listed in Table 4.

4.3 Washout Scenario P_0

In the washout scenario, the numerical solution converged to

$$(S, x, y, z) \approx (1, 0, 0, 0),$$

Main chemostat simulation scenarios

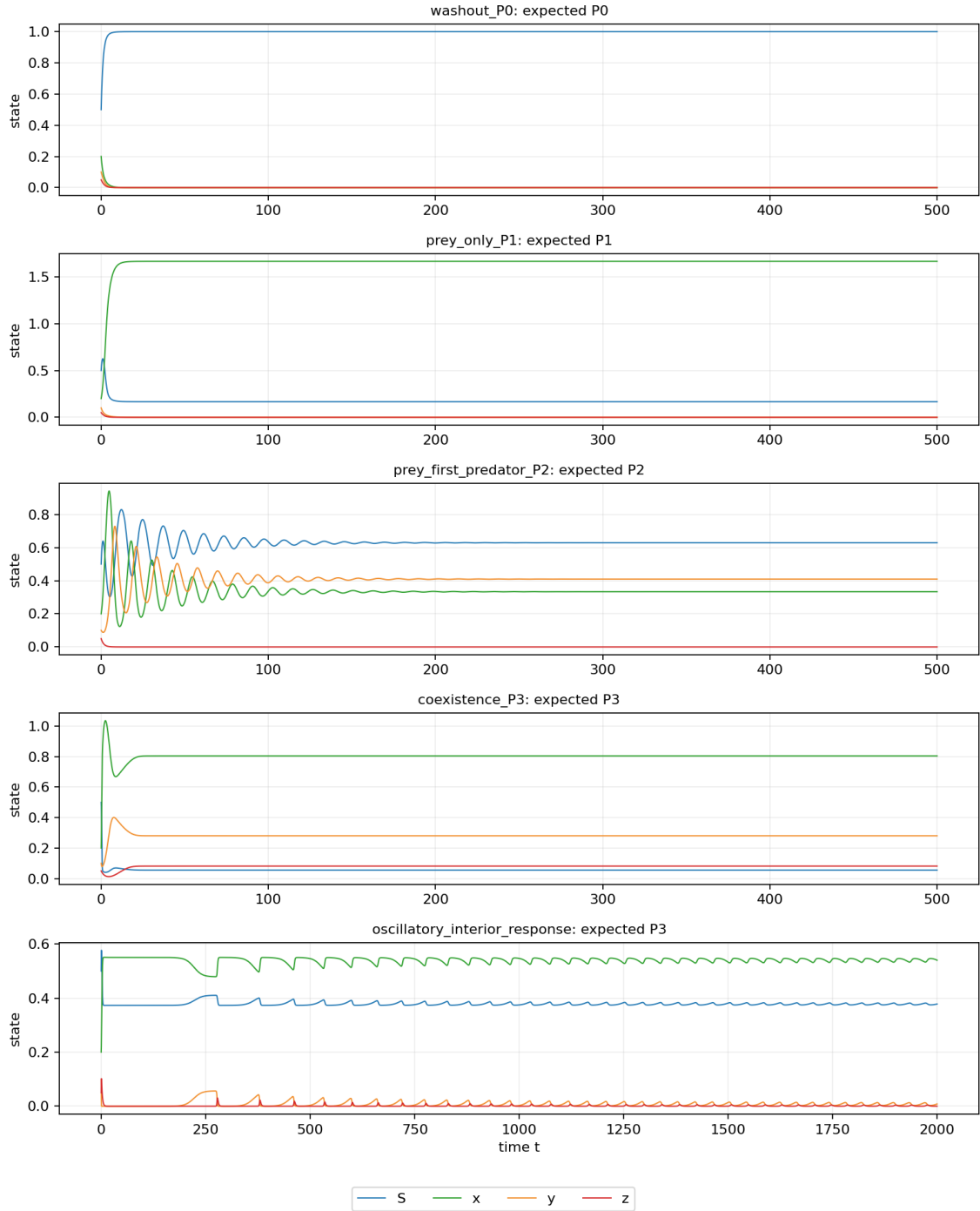


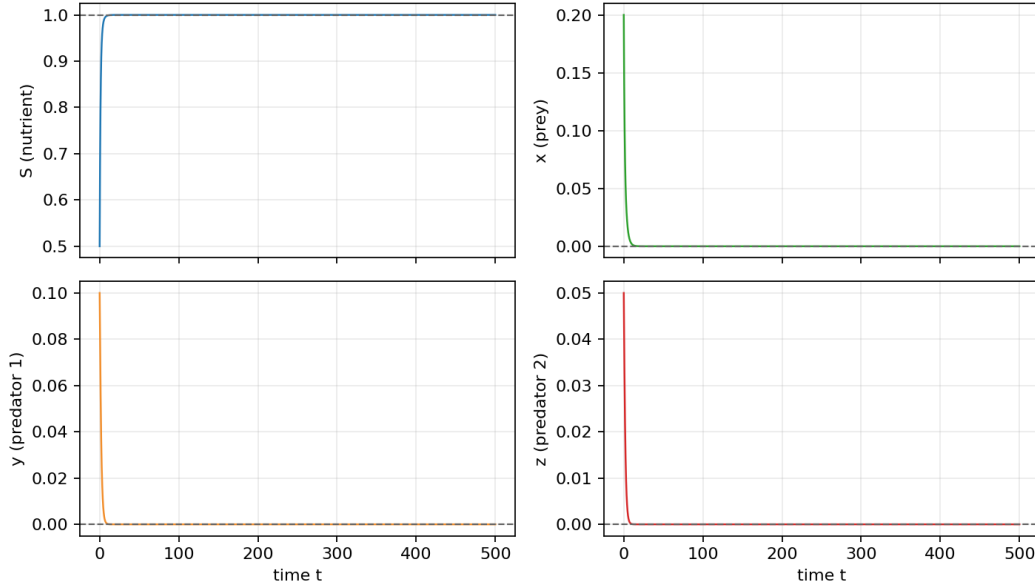
Figure 1: Overview of the five main simulation regimes generated by the RK4 simulator.

Table 4: Parameter sets used in the main simulation experiments.

Scenario	a_1	k_1	a_2	k_2	a_3	k_3	D_1	D_2	D_3
P_0 washout	0.60	0.50	1.50	0.50	1.20	0.40	0.80	0.70	0.70
P_1 prey only	2.00	0.50	0.50	1.00	1.00	1.00	0.50	0.60	0.60
P_2 two species	2.00	0.50	2.00	1.00	0.50	1.00	0.50	0.50	0.70
P_3 coexistence	4.80	0.17	4.30	1.85	4.60	1.66	0.72	1.11	0.665
Oscillatory response	6.52	1.76	5.07	2.64	3.78	0.0466	1.14	0.778	0.488

with distance 5.55×10^{-15} from the analytical equilibrium P_0 . Biologically, this means that the prey cannot grow fast enough on the incoming nutrient to overcome its removal rate. Once the prey disappears, both predator populations also disappear because their food sources are absent. The nutrient therefore returns to its normalized input concentration.

washout_P0: Complete organism washout
(dashed line: theoretical equilibrium)

Figure 2: Running simulation image for the washout scenario P_0 .

4.4 Prey-Only Scenario P1

In the prey-only scenario, the final numerical state was

$$(S, x, y, z) \approx (0.166667, 1.666667, 0, 0),$$

with distance 2.57×10^{-14} from the theoretical equilibrium P_1 . The prey population survives because it can exploit the nutrient supply, but the resulting prey concentration is not sufficient for the first predator to overcome its own removal rate. Consequently both predators wash out. The nutrient level is much lower than in P_0 , showing that prey persistence is associated with continuous nutrient consumption.

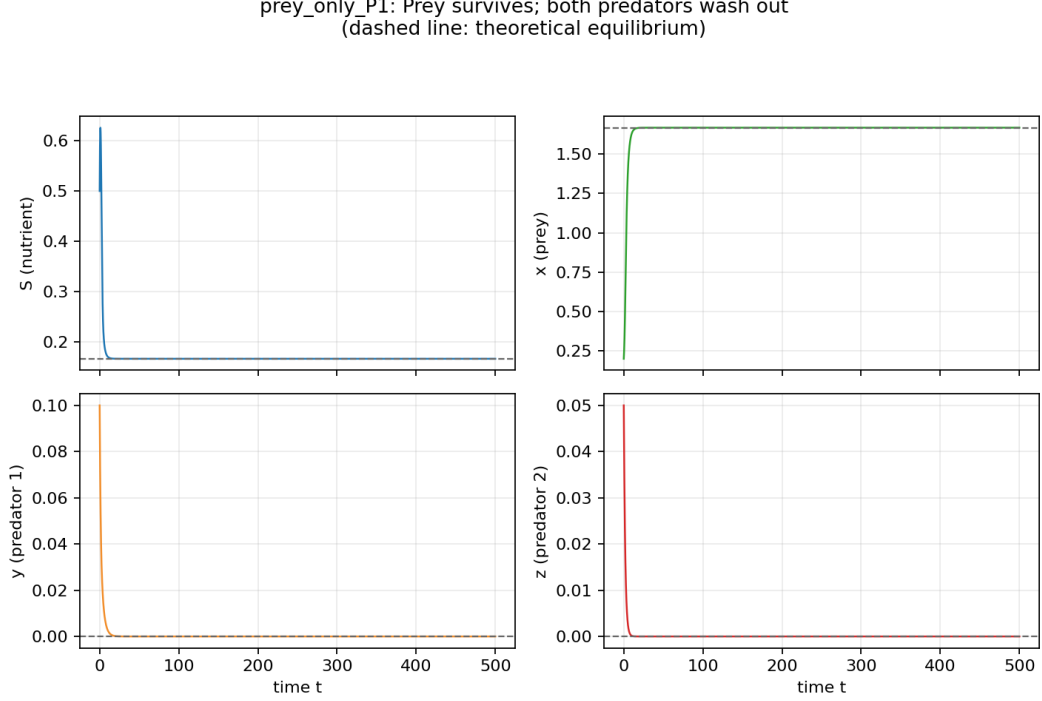


Figure 3: Running simulation image for the prey-only scenario P_1 .

4.5 Two-Species Scenario P2

In the two-species scenario, the solution approached

$$(S, x, y, z) \approx (0.628666, 0.333334, 0.409333, 0),$$

which is 1.29×10^{-6} from the analytical equilibrium P_2 . The prey and first predator persist, while the top predator is removed from the system. This confirms the expected intermediate regime in which the first predator can invade the prey-only equilibrium, but the second predator cannot invade the prey-predator equilibrium. The approach to P_2 contains small damped transients, so temporary oscillation should not be confused with sustained periodic behavior.

4.6 Coexistence Scenario P3

In the coexistence scenario, all state variables converged to positive values:

$$(S, x, y, z) \approx (0.055067, 0.804597, 0.280534, 0.081550).$$

The distance from the analytical coexistence equilibrium P_3 was 1.94×10^{-11} . This close agreement shows that the simulator reproduces the expected interior equilibrium for the selected parameter set. In biological terms, nutrient enters the chemostat continuously, the prey consumes nutrient, the first predator consumes prey, and the top predator consumes the first predator. This numerical result supports the persistence interpretation in the reference, although a single trajectory is not a proof of uniform persistence for all admissible positive initial conditions.

4.7 Oscillatory and Damped-Oscillatory Response

The fifth scenario was included to demonstrate an interior response with visible oscillations. Its analytical target equilibrium is

$$P_3 = (0.377311, 0.540994, 0.006908, 0.001193).$$

prey_first_predator_P2: Prey and first predator survive
(dashed line: theoretical equilibrium)

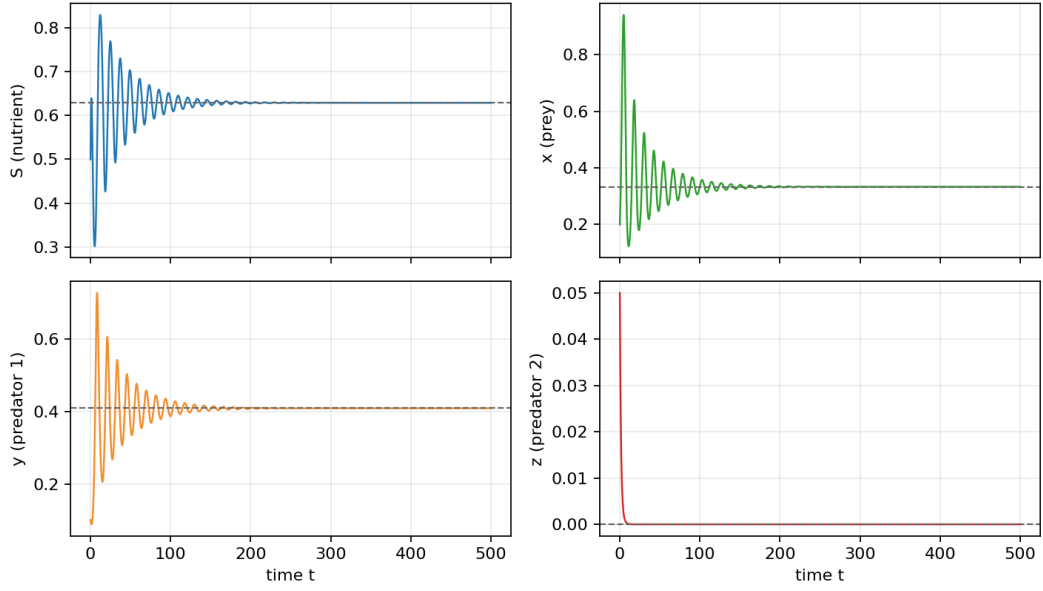


Figure 4: Running simulation image for the two-species scenario P_2 .

coexistence_P3: All trophic levels coexist
(dashed line: theoretical equilibrium)

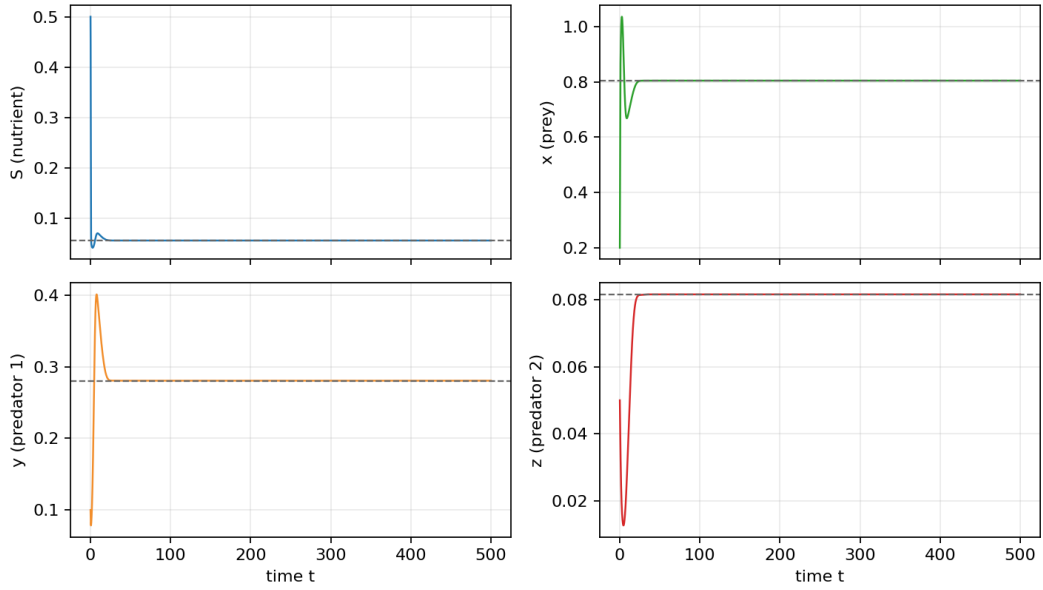


Figure 5: Running simulation image for the coexistence scenario P_3 .

At $t = 2000$, the instantaneous numerical state was

$$(S, x, y, z) \approx (0.377832, 0.539609, 0.008618, 0.000146),$$

with distance 2.49×10^{-3} from P_3 . This final-state distance is larger than in the previous cases because the endpoint depends on the phase of the remaining oscillation.

To avoid misclassification, the oscillatory case was also evaluated using window statistics. Over the late window $1500 \leq t \leq 2000$, the mean state was only 4.66×10^{-4} from P_3 . After $t = 500$, the trajectory had 23 detected local maxima in y and 20 detected local maxima in z . The maximum ratio between the late-window amplitude and the middle-window amplitude was 0.425, so the oscillations were decreasing. These measurements identify the behavior as a slowly damped oscillatory approach to the interior equilibrium. They are consistent with dynamics near a Hopf stability boundary, but they do not by themselves prove a Hopf bifurcation.

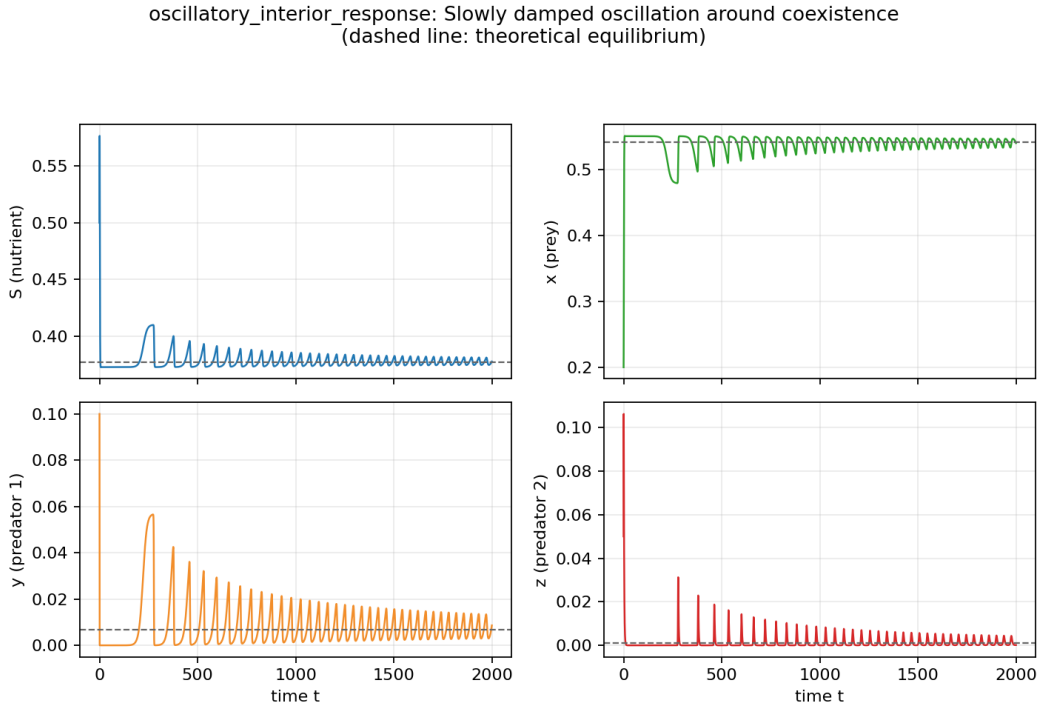


Figure 6: Running simulation image for the slowly damped oscillatory scenario.

4.8 Comparison Between Numerical and Theoretical Equilibria

Table 5 compares the numerical final states with their theoretical targets. The first four experiments agree with their expected equilibria to high accuracy. The oscillatory case is naturally less well represented by a single final point, so its window average was used as the main evidence for convergence toward P_3 .

4.9 Biological Interpretation of the Results

The simulations show a clear trophic survival hierarchy. In P_0 , the prey cannot overcome removal, so the entire biological chain washes out. In P_1 , the prey survives but does not create enough food supply for predators. In P_2 , the first predator persists, but the top predator cannot invade. In P_3 , all trophic levels have sufficient effective growth relative to their removal rates, so the system settles at a positive coexistence equilibrium.

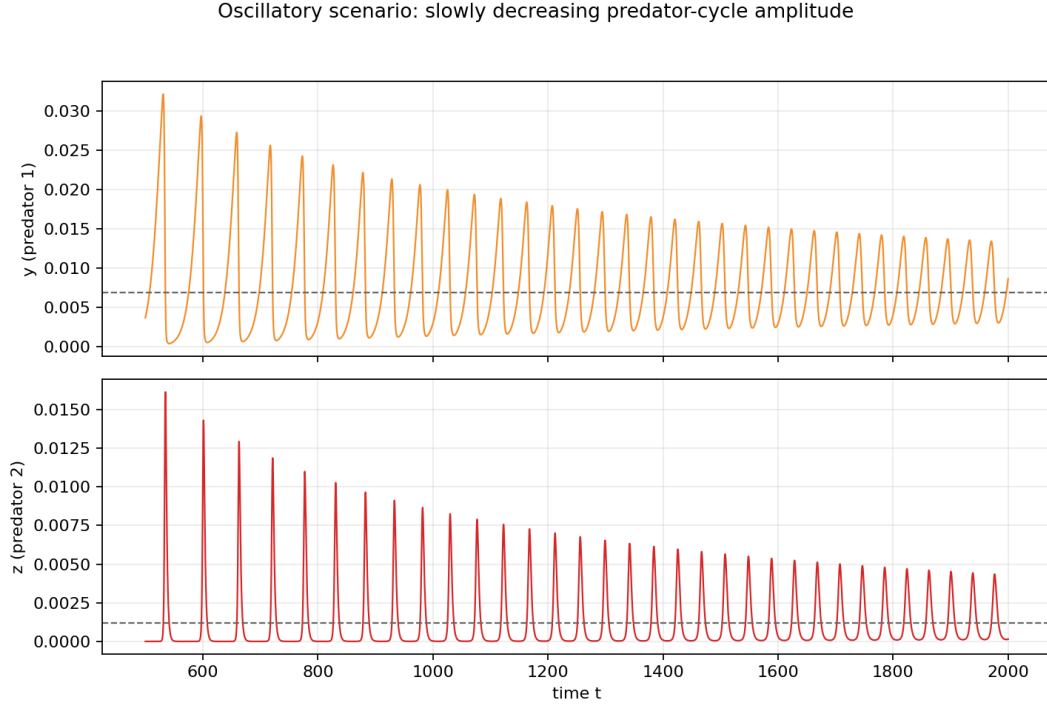


Figure 7: Zoomed view of the predator oscillations after the initial transient.

Table 5: Comparison between numerical final states and theoretical equilibria.

Scenario	Target	Numerical final state (S, x, y, z)	Distance
P_0 washout	P_0	$(1.000000, 0, 0, 0)$	5.55×10^{-15}
P_1 prey only	P_1	$(0.166667, 1.666667, 0, 0)$	2.57×10^{-14}
P_2 two species	P_2	$(0.628666, 0.333334, 0.409333, 0)$	1.29×10^{-6}
P_3 coexistence	P_3	$(0.055067, 0.804597, 0.280534, 0.081550)$	1.94×10^{-11}
Oscillatory response	P_3	$(0.377832, 0.539609, 0.008618, 0.000146)$	2.49×10^{-3}

The main biological mechanism is therefore the balance between food-dependent growth and species-specific removal. Each higher trophic level can persist only if the lower level maintains a concentration high enough to make the corresponding Monod response exceed the removal rate. The numerical results also show why long simulations and analytical equilibria should be used together. Graphs reveal transient oscillations, while equilibrium distances give a quantitative test of the predicted long-term regime.

4.10 Interactive Simulator Demonstration

The project also includes an interactive browser simulator in the `demo/` directory. It can be launched from the repository root by running

```
python -m demo.app
```

and then opening `http://127.0.0.1:8000/` in a browser. The interface allows the user to choose scenario presets, change the removal rates D_1, D_2, D_3 , adjust the Monod parameters, and observe the RK4 trajectory in real time. It also displays analytical equilibria, invasion thresholds, local stability information, and the nearest equilibrium to the current live state. This makes the simulation results easier to interpret during the presentation because the audience can see how changing a removal rate moves the system between washout, partial survival, and coexistence.

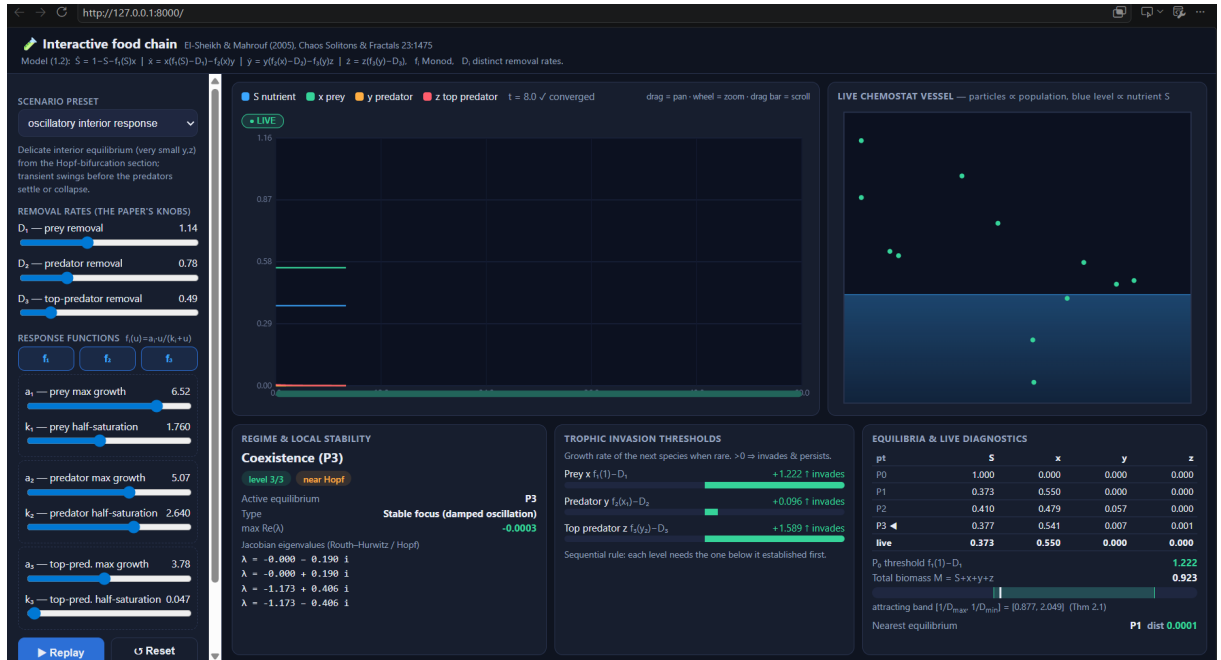


Figure 8: Screenshot of the interactive simulator showing live trajectories, parameter sliders, equilibrium table, and stability diagnostics.